Chapter 1

The Map and the Territory

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The Questions

1. What are the essential components of quantitative investing?
2. What types of securities are involved and how are they traded?
3. Who are the main market participants and what roles do they play?
4. Where do excess returns in investing come from?
5. What are the key elements that form the analytical framework of a quantitative portfolio manager?

This chapter is a guide to the essential components of quantitative investing. When considering the meaning of a word, it's often instructive to go back to its etymology, so let's play this game. Despite being a

Germanic language, English adopted many words from Latin, sometimes by way of French. "Investing" comes from "Investire", which in Latin meant "to cover with a vest", or "to put in a vest". So it should be hardly surprising that two thousand years later, vests would become the favorite garment of hedge fund managers. In the Middle Ages, the verb took on the additional meaning "to surround, to have ownership of". It is also possible that the modern meaning overtook the old because, in ceremonies in which ownership was transferred, the new owner was "invested" with a cloak and other regalia. In Italian—the direct successor to Latin—the old meaning is gone, and "investire" only means "to receive possession of something". As for "quantitative", that is Latin too: "quantum", a noun denoting something that can be measured, increased, and decreased. We will deal with ownership, sold and bought in units that can be measured, increased and decreased. This is, unfortunately, the whole of finance. You can own a house, a painting, a bet on the survival of humankind, or even an idea. Each one of these investment topics deserves its own book, written by a competent author. In writing this book, I have chosen to trade off generality in favor of detail. I have covered each subject with the goal in mind that you would have sufficient information to understand it, implement it, and critique it. However, even an analytical book needs an introduction that puts things in their proper context. In this chapter, I aim to provide that context. You will have a broad understanding of the classes of securities to which these methods apply; and of the way these securities are traded, and by whom. This is a necessary prerequisite to explore fundamental questions: Where are excess returns coming from? What causes these trading opportunities? Finally, I will present the essential components that make up the analytical framework of a quantitative portfolio manager. The underlying message is that to be successful, an investor must understand how things work. A seminal early book on investing is titled "The Intelligent Investor" (Graham, 2006). To double down on Latin, the original meaning of "intelligent" is "to read into something", similar to "insightful" in the English language. Your success will come from reasoning about the behavior of your counterparties, the rules governing the trading of your assets, and the functioning of exchanges. Many budding quants focus on quantitative methods. The fact is that theory is cheap and is often not hard. What is hard is putting the right tool at the service of the right insight.

Finally, this is the only chapter without mathematics. You should enjoy it while it lasts.

1.1 The Securities

We will be concerned with *standardized products* that are *liquid*. We explain these concepts in more detail.

To "own" an object is effectively to own *claims* on that object in the future. If you own a house, you can live in it or rent it out (your claim) and it is yours. This claim is not absolute, however. In most countries, the local or central government may need your property for reasons of public welfare and can require you to exchange your claim for cash at a fair price. If you own a painting, you may enjoy it in the confines of your house, but may not necessarily own its reproduction rights. If you own a bet on the future of humanity, your counterparty may have some *force majeure* clauses that prevent it from paying (e.g., consider a zombie apocalypse scenario). Defining ownership of an "idea" is especially challenging and prone to be treated on an *ad hoc* basis. Compared to the infinite and ever-changing nature of the meaning of property rights, our coverage is very narrow. Specifically, we focus on the subset of contracts that are standardized and liquid. We buy and sell *standardized* claims. These claims come in a few varieties, and their attributes are clearly defined and known to all potential buyers and sellers. Examples are:

- *Equities and Exchange-Traded Funds (ETFs)*. These give us partial ownership in companies, or groups of companies, and entitle us to receive future cash payments generated by the economic activities of these companies.
- *Futures*. These contracts deliver a physical commodity or a cash payment contingent on the state of the world at a future date, at a price determined today.
- *Bonds*. These are contracts that allow the transfer of debt claims among parties. An investor lends money to a borrower, in exchange for a fixed cash flow in the future (e.g., periodic interest payments and a final payment). A bond makes this claim transferable to other lenders.
- *Vanilla options*. These are claims that depend on the future value of some underlying asset; for example, you may receive the right (but not the

obligation) to buy a stock at a future date, at a price determined today. The nature of these claims is standardized, hence the term "vanilla".
- *Interest Rate Swaps (IRSs)*. These contracts allow the exchange of a certain, deterministic cash flow stream for an uncertain one, which depends on interest rates at future dates.
- *Credit Default Swaps (CDSs)*. These contracts insure the buyer against the failure of a company at a future date, in exchange for recurring fixed payments.

Further these contracts are *liquid*. For our purposes, a liquid contract is one that can be bought and sold at large enough sizes, and at sufficiently short time horizons, to enable quantitative strategies to be implemented. This means that if we plan to buy or sell a contract, we should be able to do so without incurring a transaction cost so high that our strategy is not economically attractive even for small trading sizes, and that the waiting time due to searching for a counterparty should not be so long as to make the transaction economically unattractive.

The properties of standardization and liquidity are closely intertwined. Increased standardization tends to enhance liquidity by consolidating demand, as it aggregates dispersed demand from bespoke products toward a smaller set of standardized ones. Furthermore, standardization streamlines the trading process, reducing transaction costs and simplifying contracts, thereby fostering investor trust and attracting more participants, thus enhancing liquidity. However, the downside is that customers may sacrifice the ability to trade certain useful product characteristics. Determining the optimal level of customization, even at the expense of liquidity, remains an ongoing process of learning and adaptation. For instance, prior to the 2008 financial crisis, CDSs exhibited greater variety. However, the "Big Bang" initiated by the International Swaps and Derivatives Association (ISDA) on April 8, 2009, simplified contract terms, including standardizing coupon rates (100 bps and 500 bps) and introducing a standard upfront payment, which played a pivotal role in restoring confidence in this asset class (Vause, 2010).

Trading and liquidity are at the core of the book. In order to better understand the trading process and the nature of liquidity, we should describe in some detail how trading on-exchange and over-the-counter happens.

1.2 Modes of Exchange

At any given time, economic agents want to buy or sell contracts. They want to do so quickly, securely, and cheaply. The three options around which trading is currently organized are exchanges, over-the-counter, and dark pools. Exchanges are venues in which the orders of buyers and sellers are anonymized and matched against each other. Orders are characterized by size, the number of contracts, direction (buy or sell), and price. They represent requests to buy or sell a number of contracts. The exchange records such active orders on a ledger, known as the limit-order book (LOB), and employs a set of priority rules to match buy and sell orders in the exchanges—aptly named a matching engine. In order to trade on an exchange, one must be a member of that exchange. Membership entails apparent benefits and less-apparent responsibilities. Market participants must maintain sound governance, risk processes, and capital structure.

Exchanges evolve continuously due to two driving forces. On one side, there is a push toward consolidation, which reduces operating costs and gives the owner pricing power. On the other side, technical and process innovations introduce new competitors into the market. In the United States alone, there are more than a dozen equity stock exchanges. Exchange-traded assets, such as stocks, options, and future contracts (including Forex Futures), are often liquid, although this condition is neither necessary nor sufficient: some exchange-traded assets are traded in minimal volumes and, therefore, are not liquid, and some very liquid products are traded off-exchange.

Other assets are not traded on exchanges, but *over-the-counter (OTC)*. In this case, the buyer or seller transacts through an institutional market participant, the broker-dealer, which is connected to other broker-dealers and facilitates the matching of orders. Bonds, IRSs, Forex spot currencies, Forex Forwards, and CDSs are examples of contracts traded OTC. Some of these, like currencies, are among the world's most liquid contracts. A precondition for liquidity is standardization. Think of a house. "The New York housing market" is very different from the stock market in that each of the 15.4 billion outstanding Apple shares (as of July 2024) is indistinguishable from the other and sells in a matter of seconds. In contrast, a house has many attributes that make it unique: location, size, age, blueprint, and condition. Another characteristic of liquid markets is the large number of participants. When numerous participants are involved in

a market, competing for a relatively low number of contracts, transactions become more frequent, and the necessity for bilateral bargaining diminishes. The ability of any individual participant to influence the price is significantly reduced. To illustrate, consider the housing market as a counterpoint: when selling a house, you typically negotiate with one specific buyer (out of a few eligible ones), who may spend many hours searching for the right property and may engage in intense bargaining, sometimes to the point of contention, to secure the best possible price.

Finally, *Dark Pools* (a type of ATS, or Alternative Trading System, which does not make its LOB transparent) are additional venues that are distinct from exchanges (although sometimes owned by them). Dark Pools address the needs of certain institutional investors to execute orders without displaying their trading intentions. By design, Dark Pools hide order details and only make trades details available after execution. As of 2024, approximately 16% of U.S. shares are traded on Dark Pools.

1.3 Who Are the Market Participants?

It is convention to partition traders into the *sell side* and the *buy side*. The former facilitates trading by providing services; the latter receives trade for their own benefit. Below I describe the participant types. For a more detailed description, see Harris (2003).

1.3.1 The Sell Side

The sell side comprises brokers, dealers, and broker-dealers.

- *Dealers*[1] fulfill their clients' demands, thus providing liquidity. They take the opposite side of the trade; they are profitable if, on average, they sell (buy) at a price higher (lower) than what they initially paid to buy (sell) the asset. The difference between buy and sell prices is the *spread*. When dealers interact with clients, they quote the buy price (the *bid price*) and the sell price (the *ask price*) for a contract. They are effectively *making a market*, since these quotes make transactions possible. In OTC markets, dealers are the primary liquidity providers. The most sophisticated among such markets allow the dealers to quote prices,

[1] Ch. 13 in Harris (2003).

quantities, and other attributes continuously. For example, fixed-income products can be traded on Dealerweb or Bloomberg. In order markets or for highly bespoke products, the dealers quote on request, possibly one-sided only, for a specific quantity and with an expiration time. The quote, or the spread if the quote is two-sided, depends on the quantity. Similarly to speculators, dealers trade on their own behalf. Like speculators, they hold a portfolio (or an *inventory* of positions) and face the issue of trading counterparties that may be more informed than themselves. Unlike speculators, dealers are passive traders, in that they respond to their clients. Also, unlike speculators, dealers enjoy special regulatory status. Because dealers observe the demand flow of their clients, they are informed agents, often serving the needs of informed clients. The dealers' profit originates from the realized spread of their trade (which is usually lower than the quoted spread) but also from the specific information the dealer derives from the order flow. One specific type of flow originates from retail investors (who we introduce later in the chapter). These investors access the market indirectly through brokers. Brokers have special arrangements to direct market orders to dealers, who commit to executing them while offering certain price guarantees on the trade.

In summary, dealers are liquidity providers, and they are compensated for services through trading profits.

- *Brokers*[2] trade on behalf of their clients. When the broker receives an order from a client, together with information about the client's time and price preferences, it searches for the most effective channel to execute it in accordance with these preferences. For example, a client sends a broker an order to buy a certain number of shares of a company. The broker is a member of all major exchanges. It splits the large order into smaller orders and routes them to the various markets at times that meet the execution horizon of the client or its expected cost. Unlike dealers, brokers are intermediaries who take no risk by holding contract positions at any given time. The intermediation service they provide is beneficial; however, it comes with its own risks. First, brokers provide exchange access to non-member clients and they provide OTC dealer access to non-institutional clients. Institutional clients, too, may want to enlist a broker when interacting with dealers, since the broker

[2] Ch. 7 in Harris (2003).

anonymizes the clients. Further, broker intermediation solves bilateral settlement risk: money is exchanged for contracts after the trade occurs. Clients need to know and trust their counterparty to protect themselves from insolvency, reneging, or non-compliance. There is a small number of brokers compared to the number of traders, so that clients need to approve (and be approved by) only a few counterparties; a reduction in time, cost, and risk. This, of course, does not eliminate counterparty risk. It transfers it to the brokers. The brokers manage it by vetting the clients, and by requiring that clients deposit capital at the broker, which the broker uses in case of client insolvency. The brokers also *clear* and *settle* trades on behalf of the client. In addition to these services, brokers, and especially *prime brokers*, the subclass of brokers servicing hedge funds and other sophisticated investors, offer their clients other services:

- *Custodial services*. Brokers ensure receipt, recording, and safekeeping of securities.
- *Rehypothecation*. Clients may allow the brokers to use their securities for the brokers' own needs in exchange for fees or rebates. For example, brokers may use client securities as collateral for their own transaction or lend them to other clients.
- *Margin loans*. Brokers lend clients short-term capital to buy securities. They charge them SOFR[3] plus a spread.
- *Location of short positions*. Clients may want to *short* stocks (i.e., sell shares first, and buy them back at a later time), with the expectation that future prices will be lower than current ones. Brokers enable these transactions by lending shares from a third party and making them available to clients. The clients then sell them in the open market, buy them back at a later time, and return them to the broker. After the initial sale, but before the buyback, the broker invests the cash proceeds at SOFR plus a spread. The client receives from the broker SOFR minus a spread.
- *Research reports and services*, as well as broker-specific data. These services used to be bundled in broker commissions but after the implementation of MIFID II regulation, they are now charged separately.

[3] Secured Overnight Financing Rate (SOFR) is a measure of the interest rate charged for overnight cash loans.

– *Capital introductions*, in which brokers facilitate the connection between hedge funds and potential investors.

In summary, brokers offer diversified services, the most important of which is to facilitate clients' transactions. They are compensated by commissions, interest on cash balances, interest on lending, and *payment for order flow* (PFOF), which is the compensation brokers receive from market makers for routing orders to them.

Broker-dealers, also called *dual traders*, combine the previous two functions in a single entity. They act both on behalf of the client and on their own behalf. This introduces a tension. The dealer's arm is incentivized to use the broker's information in trading to its advantage. Maybe the simplest action is *front-running*: the dealer is aware of incoming buying or selling demand for a security, and buys it in advance before this demand manifests in the market and is reflected in prices. To mitigate this type of behavior, regulations are in place to safeguard the interest of the client. The most important law regulating brokers, dealers, and broker-dealers is the Securities Exchange Act of 1934 (or "1934 Act").

1.3.2 The Buy Side

The buy side usually trades with the sell side. You (the reader of this book) are likely to be a member of this group, even though certain dealers face quantitative challenges similar to yours. It is important to understand who the actors in the buy-side drama are, because you will continuously interact with them, and your excess returns will be the outcome of this interaction. We could classify the buy-side actors according to several criteria. For example, the sub-industry to which they belong: life insurers, mutual funds, hedge funds, and so on. I opt to classify them (subjectively!) based on the type of investing they perform.

- *Indexers* are passive investors. Their portfolios replicate the compositions of the benchmarks, or indices, generated by data providers like MSCI, S&P, Russell, CRSP, or from exchanges like FTSE 100, TOPIX, and Deutsche Börse. These indices are updated on a quarterly or biannual basis, and they comprise bond indices as well, like the Bloomberg Agg (until 2016 owned by Barclays). Several investment vehicles track indices; mutual funds and exchange-traded funds are the largest in terms of size. Large firms in this group are Blackrock, Vanguard, and State

Street. Indexers make up a large and growing share of the total asset base. According to estimates by Chinco and Sammon (2023), they represent over 37% of the U.S. stock market capitalization as of 2020.

- *Hedgers* are firms participating in markets with the primary objective of reducing financial risk originating from their core businesses. For example, currency risk is faced by any firm doing business internationally. Firms such as airlines and manufacturing companies purchase fossil fuels (gas, Brent, and West Texas Intermediate), whose price variability can be very disruptive. Hedgers primarily participate in derivative markets: futures, swaps, and options. Hedgers differ from other participants who also hedge, such as dealers or hedge funds, in that hedging is the primary activity they perform.
- *Institutional active managers* are firms investing on behalf of their clients. They run strategies that are sometimes benchmarked to commercial indices and hope to beat them. There is some evidence of underperformance of funds serving retail investors; see S&P SPIVA report[4] or the Refinitiv study (Glow, 2023). Both show that over 60% of funds underperform their benchmarks over a one-year trailing basis. The outperformance of funds over one year is not persistent: as of January 2024, 91% of funds trail the performance of the S&P500 over the previous 15 years. On the other side, funds serving institutions seem to beat their benchmarks (Gerakos and Linnainmaa, 2021). The *tracking error* is a measure of the risk they can take when differing from their reference benchmarks. Otherwise stated, their portfolios can be expressed as the sum of the positions in a benchmark, and of discretionary positions of a "tracking portfolio". A large tracking error gives the funds much discretion; a low one places their returns close to the indices, and makes them "index huggers," or "closet indexers" (i.e., index funds in disguise).
- *Asset allocators* manage portfolios composed of securities in multiple asset classes. Within an asset class, the portfolio closely follows a representative benchmark. One can view asset allocators as managers of a portfolio of asset classes. The relative weight of these asset classes in the portfolio is either constant or changes slowly. Common asset classes are equities, bonds, commodities, and cash equivalents.[5] In addition, asset allocators

[4] https://www.spglobal.com/spdji/en/research-insights/spiva/.

[5] *Cash equivalents* are highly liquid assets with low returns, like bank certificates of deposit, short-dated treasuries, or commercial paper.

invest in alternative asset managers like private equity firms, venture capital, hedge funds, and real estate.

- *Informed traders* include primarily hedge funds and principal trading firms. These firms are usually organized as partnerships, although a few are public companies. They face fewer constraints than institutional managers. Whereas principal trading firms only have general partners (GPs, the principals) investing their own money, hedge funds also have limited partners (LPs) who do not invest actively.[6] These firms pursue absolute returns (i.e., not tracking a benchmark), which exhibit low correlation to the indices of major asset classes.[7] Informed traders invest heavily in human capital, technology, and data to achieve this goal. They fulfill two major functions. The first one is *price discovery*. By using all information available to them, they generate estimates of the true value of securities. If the security prices differ from their estimates, they trade to exploit the mispricing. If the price is lower than their estimate, they buy the security. In the process, they increase its price and bring it closer to equilibrium. Mispricing can take many forms. If the same security is offered at different prices on different exchanges, *arbitrageurs* (a subset of informed traders) will try to exploit the difference; of course, this may not be easy to do, so the difference either persists, or disappears very quickly due to technology investment in low-latency trading. The second role of informed traders is *liquidity provisioning*. Supply and demand of certain assets is predictable to a certain degree. I provide examples in Section 1.4; hedge funds and market makers develop specialized strategies that predict imbalances, hold (or short) securities before the liquidity need materializes, and meet the liquidity needs at the event. The range of possible intervals between prediction and event can be vast—from sub-second for high-frequency market makers to weeks or months for hedge funds.
- *Retail investors* trade for their own account via retail brokers. In 2020, retail investors made up approximately 20% of total volume; the share

[6] GPs in hedge funds are both principals (since they invest their own money) and agents (on behalf of the LPs). To resolve this conflict of interest, the SEC regulates the class of investors who can be LPs (high-net-worth, sophisticated investors), and gives LPs special privileges within the fund.

[7] This is not always true in practice because (a) some hedge funds have an explicit market exposure and (b) some hedge funds have an asymmetric exposure to the market (Agarwal and Naik, 2004).

was slightly more than 10% in 2011. Several studies, across different national markets and periods, have shown that retail traders are consistently unprofitable (Barber and Odean, 2013); retail trader flow is uninformed. This is one of the reasons why it is highly sought after by dealers, who will pay the retail brokers for routing it to them (payment for order flow).

1.4 Where Do Excess Returns Come From?

Now that we have introduced the main actors in the play (usually a tragedy, rarely a comedy, and occasionally a farce) of investing, we can discuss the sources of excess returns. The "excess" qualifier means "in excess of portfolio invested in risk-free assets, such as short-dated U.S. treasuries." This topic is central both to academic financial research and to practitioners. Academic finance is primarily concerned with the question of *efficiency*. In the words of Malkiel (1987):

> *A capital market is said to be efficient if it fully and correctly reveals all available information in determining security prices. Formally, the market is said to be efficient with respect to some information set, ϕ, if security prices would be unaffected by revealing that information to all participants. Moreover, efficiency with respect to an information set, ϕ, implies that it is impossible to make economic profits by trading on the basis of ϕ.*

An exceptionally concise definition, if there ever was one. At its core is the "information set ϕ". This could be, for example, the set of all historical prices of the traded securities. Nowadays, this information can be obtained with *relatively*[8] little effort. A different type of information set is publicly available information,[9] defined in the United States as "any information that you reasonably believe is lawfully made available to the general public from: (i) federal, state, or local government records; (ii) widely distributed media; or (iii) disclosures to the general public that are required to be made by federal, state, or local law." An even finer information set

[8] The "relatively" here denotes a bit of sarcasm. Collecting long time series of good quality, accounting for corporate actions, is hard, expensive, and requires skill.

[9] PRIVACY OF CONSUMER FINANCIAL INFORMATION UNDER TITLE V OF THE GRAMM–LEACH–BLILEY ACT. §163.3 (w) (1).

is the set of *all* information available to *any* investor. Academic research tries to determine the validity of the statement that "security prices would be unaffected by revealing that information to all participants." Note that this does not mean that ϕ is not helpful to predict future prices. Indeed, there is empirical evidence that asset prices are predictable. However, the hypothesis is that current prices may not be affected. We do not trade in the direction of returns, up to the point that the investing opportunity disappears. This is unintuitive. Why would we not take advantage of an informative prediction? One reason is *risk*. Even if we have some information about the future return of an asset, the uncertainty around the prediction is too high for us to take advantage of it. For example, say that, to the best of our knowledge of ϕ, we expect the U.S. market to appreciate 8% next year, while our cash custodied at the broker will return a measly 2%. Does this imply that we will rebalance our portfolio to 100% a market-tracking asset like SPY? Hardly. The reason is that the standard deviation of market returns is 20%, a little too high for comfort. Risk, however, is not the only reason. Another one is *liquidity*. Indeed, the road to hell of an investor is littered with quite accurate predictions of assets that barely trade or do not trade at all. A famous example is the spin-off of Palm (a now-defunct mobile device company) by 3Com (a telecom equipment maker, also defunct) in 2000. 3Com floated on the public market 5% of the shares of Palm, while retaining the other 95%. Right after the initial public offering (IPO), Palm had a market value of $54B, while 3Com had a market capitalization of $28B. The implied value of 3Com assets was –$22B, even though the company had no debt, $1B in cash, and positive cash flow. Either 3Com was dramatically undervalued, or Palm was dramatically overvalued. An investor could have therefore bought 3Com shares and shorted Palm for an equal amount. The portfolio comprising these two assets was a synthetic asset whose return could be predicted. There was a problem, however. Palm shares were in short (pun intended) supply. In order to short a share, the investor must first borrow it, at a rate decided by the lender. If quoted at all, these rates were so high as to make the trade either unattractive or impossible.

Risk and liquidity are not the only two factors limiting the exploitation of information. We list three more. The first one is *funding*. Consider a scenario in which certain assets, or certain portfolios,[10] have lost much

[10] As in the case of the pair 3Com–Palm, we can interpret portfolios as synthetic assets.

of their value due to market distress. We are managing a small hedge fund, which has also lost money in this environment. Based on historical examples, we have a strong belief that such assets will rebound. Such scenarios occur quite regularly, especially in "deleveraging spirals". However, we do not have much capital available to post as margin. In addition, we need a capital buffer in order to withstand a possible additional loss in the very short term. Funding constraints prevent us from buying the asset, in spite of our accurate forecast.

A significant source of excess returns arises from *flow predictability*. Some agents, notably institutional investors and market makers, but not only these, will trade known securities on known dates. Speculators can then take advantage of this information by providing liquidity beforehand.[11] One of the most important instances of this is *index rebalancing*. Several index providers update the weights of their indices on predetermined dates using well-defined rules. Some securities are added to the index, others are removed, and finally most of the remaining ones have an updated weight. The term used for this process is *index reconstitution*. For example, TSLA was added to the NASDAQ 100, effective July 15, 2013. The announcement was made on July 10, 2013; but several investors could have forecasted the event well before that date. These investors would then purchase TSLA shares and sell them at the closing auction of July 15, 2013. The ETF, mutual funds, and bespoke products that track the index have an obligation to buy TSLA on the close of that day, and the resulting demand is likely to push up the stock price. The informed investors providing liquidity do not do so risk-free. They hold the stocks until the effective date, and over these days are exposed to the risk that TSLA may suffer from company-specific or industry-specific losses. Moreover, there is the remote risk that the reconstitution be cancelled or postponed. The size of passive investing is large and its estimate ranges from 17.5% (Novick, 2017) to 38% (Chinco and Sammon, 2023) of total assets under management. The buyer of index products bears the indirect cost of such rebalancing (Li, 2021). This is just one prominent example of predictable flows, but several others exist, usually smaller in size, but also not as widely known as index rebalancing. Their common feature is the existence of

[11] Note that this is not the same as *front running*. The latter also consists of buying or selling a security based on a demand forecast of said security; but in the latter case, the forecast relies on *non-public* information.

institutional or procedural constraints (sometimes driven by internal processes, other times by regulatory requirements) that introduce predictability in the demand of securities.

Finally, we consider a last source of excess returns: *informational advantage*. This means that the investors not only differ by their risk attitude, their tolerance to illiquidity, and their funding level, but also by their information sets, and some investors have better, more accurate forecasts of future prices or returns. This is what is often meant by "statistical arbitrage," the ability to predict returns accurately based on insights our competitors do not have.

In summary, even assuming accurate information owned by participants, some of them cannot exploit this information. We have listed several possible causes of return predictability:

- pure arbitrage;
- heterogeneous risk preferences;
- liquidity;
- funding;
- flow predictability;
- informational advantages.

These categories are not exclusive. For example, flow predictability and liquidity are related, albeit not identical; and the distinction between being compensated for risk-taking and holding an actual information advantage is usually unclear. However, these broad classes can help us reason about one strategy's edge.

1.5 The Elements of Quantitative Investing

The investment process is usually viewed as a highly structured process. There are separate components, the development of which is the responsibility of separate teams. In Figure 1.1, I show a possible organization of these components, which I follow in the organization of the book. I review them below.

- *Data*. The essential inputs to investing are under the "Data" section to the extreme left.

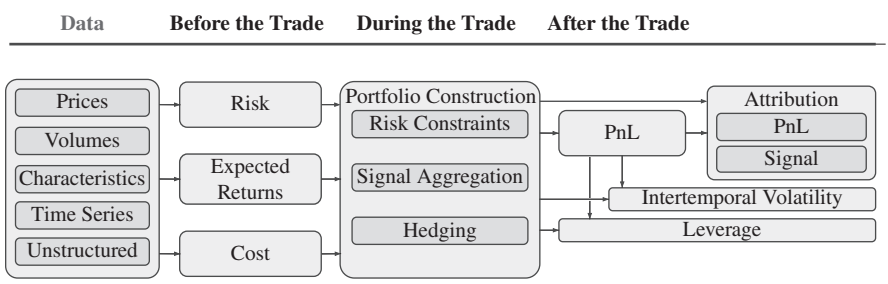


Figure 1.1 The components of the investment process.

- *Prices and trading volumes* are often collected at regular intervals (e.g., every minute, every 5, 10, 15 minutes, or daily). For high-frequency strategies it may be necessary to use order-level exchange data.
- *Characteristics* are numerical vectors associated with a security and a timestamp. Consider them as descriptors of the security. For a stock, a characteristic may be a measure of "quality", like the free cash flow generated by the firm in the most recent quarter, divided by the market capitalization of the firm. Another widely used characteristic is the realized return of the stock over a certain interval, for example the past six months.
- *Time series* differ from characteristics (which are multivariate time series) in that they are not associated with individual securities, but rather provide additional information entering the investment process. Examples of time series are the Consumer Price Index (CPI), which can be used to estimate the inflation rate in the United States; the yield of the 10-year Treasury bond; the Federal Funds Effective Rate, the overnight rate for unsecured lending of reserves among commercial banks; and the VIX, a forward-looking measure of U.S. equities market volatility.
- *Unstructured data* are the dark matter of financial data. Prices, characteristics, and times series are structured data: numerical, categorical, or ordinal (i.e., rankings), and in tabular form. Unstructured data are usually character sequences representing natural language (e.g., earnings transcripts and firm news) or images (e.g., satellite images), video/audio files, or multimodal data (i.e., a combination of all of the above formats).

- *Before the trade*. Data are used to develop three components that enter the portfolio construction during the trade. Because they precede the trading process, I classify their development as being "before the trade".
  - *Risk*. The word "risk" can mean many things. In the context of this book, we will use portfolio volatility as a proxy for risk. Estimating this volatility for an arbitrary portfolio is a challenging task.
  - *Expected returns*. In order to be profitable, a trading strategy needs to have informative predictions of future returns. This is often viewed as the paramount concern of a quantitative investor, and to a large extent it is. In nearly all modern trading systems there is more than one estimate of expected returns. The number of estimates can run into the thousands or millions.
  - *Transaction costs and market impact*. Trading securities is expensive, and these costs are unavoidable when deploying a strategy. Among other things, transaction costs determine whether a predicted return at some horizon can be turned into a profitable strategy or not; and what is the maximum profit that can be extracted from such a strategy.
- *During the trade*. The three components developed in the previous stage are combined in the portfolio construction phase. This happens "during the trade", because the portfolio construction procedure results in real-time trading decisions, and these decisions determine the Profit and Loss (PnL) of the strategy. The decisions taken in the portfolio construction process are:
  - *Incorporation of risk constraints*. A strategy's PnL is a function of the maximum risk that it can take. This is usually represented in the portfolio construction problem either in the form of constraints, or in the form of penalties added to the objective function. Risk constraints and penalties can have a very material impact on the performance of the strategy.
  - *Signal aggregation*. We use the term *signal* for a model of expected returns. As I mentioned above, there can be many signals for the returns of a single asset. A problem encountered in practice is combining such signals into a single signal.
  - *Hedging decisions*. Certain trading strategies have *exposure* to systematic risk. In layman's terms: they can lose money because their returns are correlated to market-wide sources of risk. Some of these sources can

be *hedged*, which means that such risk can be counterbalanced. This is an important concept in portfolio construction.

- *After the trade*. The trading process generates a time series of PnL, both for the overall portfolio and for its constituents.
  - *Performance attribution*. In the portfolio construction phase we estimate expected returns. This is an *ex-ante* exercise. *Ex post*, we observe the actual PnL of the portfolio. Performance attribution is the practice of tracing back performance to its possible sources, to see what worked and what did not, so that we can learn from the experience. Moreover, performance attribution can also be employed to assess whether we have skill in sizing our bets.
  - *Intertemporal volatility allocation and leverage*. How should we allocate risk across periods? This decision is very consequential to capital growth. A closely related question is that of leverage, defined as the ratio between Gross Market Value (GMV) of the portfolio and Assets under Management (AuM).

I stress that different arrangements are possible; for example, Narang (2024) employs a simpler scheme, and Pedersen (2015) has yet another one. Some of the individual elements I introduce are present in these models of quantitative investing.

The Takeaways

1. Market participants can be broadly classified into sell-side and buy-side participants. The *sell-side participants* are
   - *Dealers*
   - *Brokers*
   - *Broker-dealers*

   The *buy-side participants* are
   - *Indexers*
   - *Hedgers*
   - *Institutional active managers*
   - *Asset allocators*
   - *Informed traders (e.g., hedge funds, principal trading firms)*
   - *Retail investors*
2. *Excess returns* arise from five major sources:
   (a) *Risk*
   (b) *Liquidity*
   (c) *Funding constraints*
   (d) *Predictable flows*
   (e) *Informational advantage* in predicting future returns
3. *Quantitative investing* employs analytical models that can be partitioned into three broad groups:
   (a) *Risk measurement* models and data
   (b) *Market impact models*
   (c) *Models of expected returns*
4. The *investment process* has several stages:
   (a) *Data collection*: Prices, trading volumes, asset characteristics, time series data, and unstructured data
   (b) *Pre-trade analysis*: Risk assessment, expected returns estimation, and transaction cost analysis
   (c) *Portfolio construction*: Aggregation of signals, application of risk constraints, and hedging
   (d) *Post-trade evaluation*: Performance attribution and risk adjustments based on intertemporal volatility and leverage considerations